Adjacent Complementary Network for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Salient object detection in optical remote sensing images (RSI-SOD) is a newly developing task and has attracted extensive attention. Although some satisfactory performances have been achieved, there are still some poor performance situations, such as complex backgrounds and objects with various scales. Because the existing methods have the issue of insufficient adjacent feature representation and inappropriate constraint supervision. To alleviate the issues, we attempt to utilize the adjacent complementary and introduce the global constraints. Accordingly, we propose the adjacent complementary network (ACNet) with global constraints for RSI-SOD. Specifically, in the encoder, we mainly design the adjacent complementary module (ACM) to complement the features among the current layer and adjacent layers. Therefore, it can effectively aggregate saliency features. Besides, we have equipped a global average pooling (GAP) to enhance global information. In the decoder, we design a fusion enhancement module (FEM) to enhance the feature conversion between the coding layer and decoding layer. Finally, we employ a novel loss, which introduces global constraints and can supervise the objects with various scales. Extensive studies on two benchmark datasets demonstrate that ACNet is competitive and outperforms the 15 state-of-the-art (SOTA) approaches.**

***Index Terms*— Adjacent complementary, global constraints, optical remote sensing images (RSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) simulates the human visual mechanisms to discover the most appealing regions in an image automatically. It is a crucial preprocessing task in computer vision and is employed in many advanced tasks, such as foreground extraction [1], [2], [3], object detection [4], [5], [6], image segmentation [7], [8], [9], image compression [10], [11], [12], visual tracking [13], [14], change detection [15], [16], image style transformation [17], crowd counting [18], cognitive computing [19], and so on.

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SOD is first applied in natural scene images (NSIs), also known as NSI-SOD, and then it expands to 3-D RGB-D/T images and videos [20], [21], [22]. The NSIs are captured by cameras and mainly contain some daily objects and scenes. Recently, SOD has been introduced in optical remote sensing images (RSIs), which can play a great potential in the military, agricultural, and other optical scenes. This task is named optical RSI-SOD, which attempts to extract the most visually appealing objects in RSIs, such as buildings, islands, rivers, airplanes, and vehicles. Different from the NSIs, the optical RSIs are captured irregularly from high angles by aerial sensors and satellites, which mainly contain ground object information. In recent years, there are many satisfactory methods [23], [24], [25] proposed in NSI-SOD. However, these methods perform poorly when they are directly applied to optical RSI-SOD. Actually, there are some special issues in optical RSI-SOD, such as objects of irregular contours, objects of scale variation, objects of different types, noise interference, and cluttered background. To alleviate these problems, multi-scale feature fusion strategy [26], relational reasoning [27], edge-aware strategy [28], and other methods are proposed in optical RSI-SOD. Different from them, we concentrate on the complementary fusion of adjacent features to fully obtain saliency features, which is helpful in locating saliency regions. In addition, we have innovatively introduced global constraints, which can globally supervise objects of various scales and maintain the complete structure. Accordingly, we propose the adjacent complementary network (ACNet) with global constraints for optical RSI-SOD. The visual examples of the representative methods and our ACNet are shown in Fig. 1. We can see that the regions of our results are more accurate and the structures of our results are more complete.

Specifically, to effectively aggregate saliency features, we design the adjacent complementary module (ACM) in the encoder, which can fuse and complement the features among the current layer and adjacent layers. To enhance global information, we equip a global average pooling (GAP) in the last layer. Besides, to enhance the feature conversion between encoder and decoder, we design a fusion enhancement module (FEM). Finally, we employ a novel loss, which introduces global constraints and can globally supervise the objects with various scales. The four contributions are outlined below.

1) We propose the ACNet with global constraints for optical RSI-SOD, which can fully utilize the complementarities of adjacent features.

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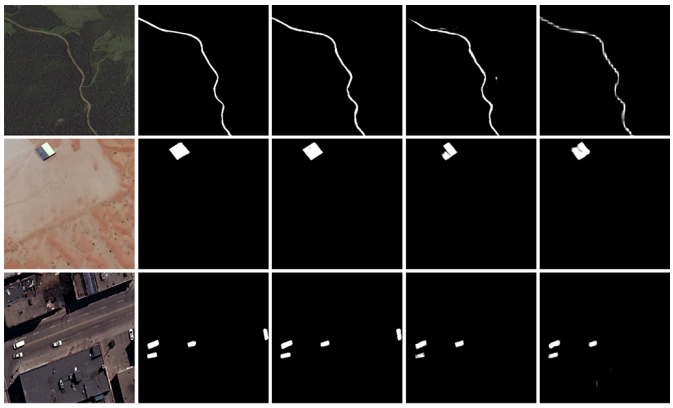


Fig. 1. Visual examples of the representative methods and our ACNet on optical RSIs. (a) Optical RSIs. (b) GT. (c) Our ACNet. (d) RRNet [27] (optical RSI-SOD method). (e) CPD [23] (NSI-SOD method).

2) In the encoder, we design the ACM to effectively aggregate saliency features, which can fuse and complement the features among the current layer and adjacent layers. In the decoder, we propose the FEM to enhance feature conversion between the coding layer and the decoding layer.
3) We employ a novel loss that adds global constraints to supervise the objects with various scales. This is the first time to introduce cosine similarity (COS) in optical RSI-SOD task.
4) We have comprehensively compared our ACNet with 15 state-of-the-art (SOTA) methods. The experimental results indicate that the ACNet is competitive and performs better.

II. Related Works

This section successively introduces methods of NSI-SOD and optical RSI-SOD.

A. Natural Scene Image-SOD

Initially, SOD was used to extract the most appealing objects in the natural scene. Itti et al. [29] developed the first visual attention model based on the difference between the center and the surroundings. Zhai and Shah [30] regarded the overall contrast of pixels as the saliency value. Achanta et al. [31] calculated the color and luminance to estimate saliency results. These methods were based on handcrafted features, which extracted low-level information from images, such as color and brightness. Because these traditional approaches cannot collect semantic information, they performed poorly in difficult conditions. The methods based on convolutional neural networks (CNN-based methods) have alleviated the issue. Many researchers explored the relationship between deep and shallow features. Hou et al. [32] presented the supervised network with a deep supervision strategy, and they introduced the short connection to enhance the relationship between deep and shallow features. Wu et al. [23] introduced a cascaded partial decoder architecture that eliminated shallow features while increasing detection efficiency. In addition, some researchers have intended to obtain accurate edges. Qin et al. [24] designed the BASNet with hybrid loss to improve boundary quality. Song et al. [25] proposed the edge refinement network with hierarchical supervision strategy. Liu et al. [33] designed the PoolNet, which explored the role of pooling techniques. Chen et al. [34] utilized global context information at different stages to alleviate feature dilution, which was useful for progressively aggregating features. NSI-SOD has achieved outstanding performance, and they provided lots of inspiration for optical RSI-SOD.

B. Optical RSI-SOD

Optical RSI-SOD was an emerging task and was facing various challenges such as complex conditions and various scales. In the beginning, some traditional methods were tried in optical RSI-SOD. Faur et al. [35] utilized the mean-shift algorithm to evaluate the prediction results in optical RSI-SOD. By analyzing the color information content, Zhang et al. [36] were able to obtain accurate saliency regions. Zhao et al. [37] used sparse representation to guide the generation of saliency results. To generate and refine the saliency results, Ma et al. [38] presented the superpixel-to-pixel approach. Huang et al. [39] adopted the online discriminant dictionary learning to generate saliency results. Besides, there were some works [40], [41], [42] for specific scenes. However, these methods can only obtain low-level features, but not semantic information. Due to the strong feature extraction abilities, CNN-based algorithms have gained excellent performance. Li et al. [43] constructed a public benchmark RSI-SOD dataset (ORSSD) with pixel-wise annotation. In addition, they proposed the LV network to perceive the various scales objects and suppress cluttered backgrounds. Zhang et al. [44] constructed a public benchmark RSI-SOD dataset (EORSSD). They proposed dense attention fluid network (DAFNet), which used low-level attention clues as a guide to enhance high-level attention maps. Liu et al. [45] proposed the global perception network, which gradually enhanced the feature representation. To better construct semantic relations and restore details, Cong et al. [27] first introduced relational reasoning in optical RSI-SOD and proposed the relational reasoning network (RRNet). To model the complementarity in local and global content, Li et al. [26] designed the multicontent complementary network (MCCNet). Tu et al. [46] extracted multiscale region features by combining boundary and region. To fully utilize context information, Li et al. [47] designed the adjacent context coordination network (ACCoNet), which used the adjacent features from the backbone. Wang et al. [48] combined the CNNs and transformer to better extract context information and multiscale features. Gao et al. [49] used the transformer to combine global and local features. For precise detection, Zhou et al. [28] employed multiscale feature fusion and introduced edge constraints. Gu et al. [50] combined the edge information and attention mechanisms to improve the applicability of the method in complex scenes. Zhou et al. [51] created the edge-guided recurrent positioning network that can maintain complete saliency regions and precise details. Some researchers have also built lightweight models. Li et al. [52]

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designed the CorrNet and adopted the coarse-to-fine strategy, which was the first lightweight network for optical RSI-SOD. Shen et al. [53] designed a lightweight network (FSMINet), which compressed high-dimensional features and introduced the multiscale strategy for adequate feature extraction. Li et al. [54] designed a novel lightweight network based on semantic matching and edge alignment, which was beneficial for practical applications.

The aforementioned methods have achieved outstanding results in optical RSI-SOD. However, they seldom consider the feature supplements among the adjacent layers, and the appropriate constraints of various scales. Different from the aforementioned methods, we consider both the adjacent features and the appropriate constraint. Specifically, we use the enhanced adjacent features, which combine the features from the backbone and the enhanced features to effectively aggregate saliency features. More importantly, we introduce a novel global constraint to supervise the objects with various scales, which is the first time to introduce COS in the optical RSI-SOD task. Therefore, we propose the ACNet with global constraints, which can fully utilize the adjacent complementary and globally supervise the objects with various scales. In Section III, we extensively introduce our ACNet.

III. Proposed ACNet

We propose the ACNet with global constraints to fully utilize the adjacent complementary and strengthen the global constraints. First, we introduce the overall structure of the ACNet, and the architecture of our ACNet is shown in Fig. 2. Then, we describe the proposed ACM and the FEM in detail. Finally, we present our hybrid supervision method, which introduces COS to supervise the accuracy of the pixels globally.

A. Network Overview

Following [55], [56], [57], we use the modified VGG-16 [58] as the encoder backbone. The resolution of the input is resized as 256×256. We use $\boldsymbol{E_i}$ to denote the corresponding module in the encoder, and their corresponding output features are f_{E_i}. Then, to improve feature expression, we mainly design the ACM, which is used to complement the features among the current layer and adjacent layers. Therefore, it can effectively aggregate saliency features and fully utilize adjacent complementary information. We use $\boldsymbol{ACM_i}$ to denote the corresponding layer of ACM, and their corresponding output features are f_{ACM_i}. Besides, we have equipped a GAP in the last layer of the encoder to enhance the global information. Finally, we design a FEM to enhance the feature conversion between encoder and decoder, which can improve the accuracy by fusing the feature maps with their corresponding positions. We use $\boldsymbol{FEM_i}$ to denote the corresponding layer of FEM, and their corresponding output features are f_{D_i}.

1) Adjacent Complementary Module: To effectively aggregate saliency features and fully utilize adjacent complementary information, we design the ACM. It is used to complement the features among the current layer and adjacent layers. The details of ACM are shown in Fig. 2. We can see that there are three inputs in ACM ($\boldsymbol{ACM_2}$, $\boldsymbol{ACM_3}$, and $\boldsymbol{ACM_4}$), and there are two inputs in ACM ($\boldsymbol{ACM_1}$ and $\boldsymbol{ACM_5}$). The output features f_{ACM_i} of the $\boldsymbol{ACM_i}$ are defined as

$$f_{\mathrm{ACM}_i} = \begin{cases} A(f_{E_i}, f_{E_{i+1}}), & i = 1 \\ A(f_{\mathrm{ACM}_{i-1}}, f_{E_i}, f_{E_{i+1}}), & i = 2, 3, 4 \\ A\left(f_{\mathrm{ACM}_{i-1}}, f^{l}_{E_i}\right), & i = 5 \end{cases} \tag{1}$$

where $A()$ represents the processing of ACM, f_{ACM_i} are the output features of the $\boldsymbol{ACM_i}$. $f_{\mathrm{ACM}_{i-1}}$ are the features of previous layer, f_{E_i} and $f_{E_{i+1}}$ are the features of current and subsequent layers, and $f^{l}_{E_i}$ is the enhanced feature from the GAP of the last layer in the encoder.

We have introduced the processing of the module in detail. First, we use parallel convolution to process the features f_{E_i} of current layer, which can fully capture multiscale information. The output features f_{P_i} are defined as

$$f_{P_i} = \mathrm{Conv}_1(\mathrm{Cat}(\mathrm{Conv}_1(f_{E_i}), \mathrm{Conv}_{3-3}(f_{E_i}), \mathrm{Conv}_{3-5}(f_{E_i}), \mathrm{Conv}_{3-7}(f_{E_i}))) \tag{2}$$

where $\mathrm{Cat}()$ means channel concatenation, $\mathrm{Conv}_1()$ means convolution with the kernel size of 1, $\mathrm{Conv}_{3-3}()$, $\mathrm{Conv}_{3-5}()$, and $\mathrm{Conv}_{3-7}()$ mean dilated convolution with the kernel size of 3 and dilated rate of 3, 5, 7, respectively.

Next, we generate spatial attention [59] through the features f_{P_i}, which can fully obtain the regional attention of the current layer. The spatial attention S_{ATT_i} is defined as

$$S_{\mathrm{ATT}_i} = \sigma(\mathrm{Conv}_3(\mathrm{Cat}(\mathrm{MaxPool}_s(f_{P_i}), \mathrm{AvgPool}_s(f_{P_i})))) \tag{3}$$

where $\mathrm{MaxPool}_s()$ and $\mathrm{AvgPool}_s()$ mean global max pooling and GAP on the channel domain to obtain spatial attention. $\mathrm{Conv}_3()$ means convolution with the kernel size of 3, and $\sigma()$ means sigmoid function.

Then, spatial attention enhances the features f_{E_i} of the current layer, the features $f_{\mathrm{ACM}_{i-1}}$ of the previous layer, and the features $f_{E_{i+1}}$ of the subsequent layer, respectively.

The enhanced features f'_{E_i} is defined as

$$f'_{E_i} = S_{\mathrm{ATT}_i} \otimes f_{E_i} \tag{4}$$

where $\otimes$ means element-wise multiplication.

The enhanced features $f'_{\mathrm{ACM}_{i-1}}$ is defined as

$$f'_{\mathrm{ACM}_{i-1}} = S_{\mathrm{ATT}_i} \otimes \mathrm{Down}(f_{\mathrm{ACM}_{i-1}}) \tag{5}$$

where $\mathrm{Down}()$ means downsampling by max pooling.

The enhanced features $f'_{E_{i+1}}$ is defined as

$$f'_{E_{i+1}} = S_{\mathrm{ATT}_i} \otimes \mathrm{Up}(f_{E_{i+1}}) \tag{6}$$

where $\mathrm{Up}()$ means upsampling by bilinear interpolation.

We aggregate these enhanced features to obtain the features f_{AGG_i}, which is defined as

$$f_{\mathrm{AGG}_i} = \mathrm{Cat}\left(f'_{\mathrm{ACM}_{i-1}}, f'_{E_i}, f'_{E_{i+1}}\right). \tag{7}$$

Next, we generate channel attention through the features f_{AGG_i}, which can filter the disturbing features. The channel attention C_{ATT_i} is defined as

$$C_{\mathrm{ATT}_i} = \sigma(\mathrm{Conv}_1(\delta(\mathrm{Conv}_1(\mathrm{AvgPool}_c(f_{\mathrm{AGG}_i}))))) \tag{8}$$

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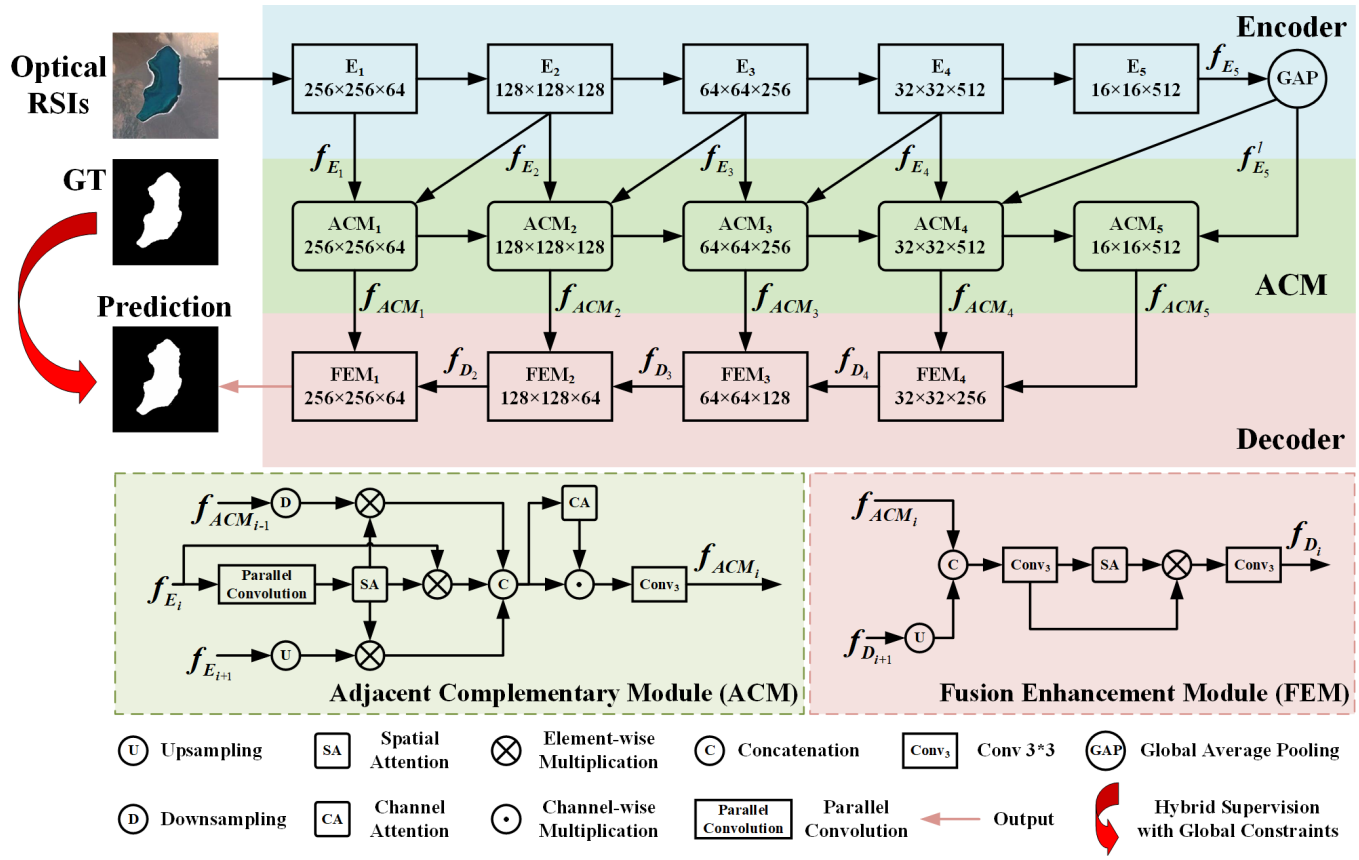


Fig. 2. Architecture of the ACNet. f_{E_i} $(i = 1, 2, 3, 4, 5)$ are the features in the encoder, f_{ACM_i} $(i = 1, 2, 3, 4)$ are the aggregation feature in the ACM, and f_{D_i} $(i = 2, 3, 4, 5)$ are the enhanced features in the decoder.

where $\text{AvgPool}_c()$ means GAP on the spatial domain to obtain channel attention. $\delta()$ means relu operation.

The enhanced features f_{ACM_i} is defined as

$$f_{\text{ACM}_i} = C_{\text{ATT}_i} \odot f_{\text{AGG}_i} \tag{9}$$

where $\odot$ means channel-wise multiplication.

From the above process, we can see that the ACM can effectively aggregate the features among the current layer and adjacent layers, it can also filter the disturbing features and obtain effective features. Therefore, it can fully utilize the adjacent complementary information.

2) Global Average Pooling: To enhance global information, we have equipped a GAP in the last layer of the encoder. The enhanced features $f^l_{E_5}$ is defined as

$$f^l_{E_5} = \sigma(\text{AvgPool}(f_{E_5})) \odot f_{E_5} \tag{10}$$

where the AvgPool() represents the GAP of multiple feature maps in space.

3) Fusion Enhancement Module: In the decoder, we propose the fusion enhancement method to enhance the feature transfer between the encoder and decoder. The details are easily lost in the long connection. To alleviate this problem, the traditional methods directly integrate the corresponding information between the encoder and the decoder. Although they can establish the information connection, there are many interferences in their fusion information, which may mislead the decoder. Therefore, we propose the FEM to enhance the feature conversion between the coding layer and the decoding layer. The integrated features are defined as

$$f'_{\boldsymbol{ADM}_i} = \text{Conv}_3(\text{Cat}(f_{\text{ACM}_i}, \text{Up}(f_{D_{i+1}}))). \tag{11}$$

Then, we generate spatial attention through the features $f'_{\boldsymbol{ADM}_i}$, which can enhance feature expression in the decoder. The spatial attention $S_{D_{\text{ATT}_i}}$ is defined as

$$S_{D_{\text{ATT}_i}} = \sigma\left(\text{Conv}_3\left(\text{Cat}\left(\text{MaxPool}_s\left(f'_{\boldsymbol{ADM}_i}\right), \text{AvgPool}_s\left(f'_{\boldsymbol{ADM}_i}\right)\right)\right)\right) \tag{12}$$

where $\text{MaxPool}_s()$ and $\text{AvgPool}_s()$ mean global max pooling and GAP to obtain spatial attention.

The output features f_{D_i} from the $\boldsymbol{FEM_i}$ are defined as

$$f_{D_i} = \text{Conv}_3\left(S_{D_{\text{ATT}_i}} \otimes f'_{\boldsymbol{ADM}_i}\right). \tag{13}$$

The FEM can enhance the feature conversion between the coding layer and the decoding layer, which is conducive to obtaining more accurate saliency results. In the training phase, we add the global constraints, which can globally supervise the objects with various scales. Section III-B describes our loss in detail.

B. Loss Function

We employ a novel hybrid loss to supervise the proposed end-to-end ACNet. The novel loss combines the binary cross entropy (BCE) loss, intersection-over-union (IoU) loss, and

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our newly introduced COS loss. Specifically, the BCE loss can supervise the accuracy of each pixel in image. The IoU loss can maintain the accuracy of the regions in image. The COS loss can maintain the consistency of the global pixels in image. In this way, the novel hybrid loss can supervise the network training under appropriate global constraints.

In SOD task, the BCE loss supervises the pixel difference between the saliency prediction and GT, which is defined as

$$L_{\text{bce}}(S,G) = -\frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} \times [G_{ij}\log(S_{ij}) + (1-G_{ij})\log(1-S_{ij})] \quad (14)$$

where S denotes prediction images, and G denotes the GT. H and W represent the height and width of the image.

Besides, the IoU loss is utilized to ensure the accuracy of the saliency prediction regions, which can supervise network training at the region level to get great results. The IoU loss is defined as

$$L_{\text{iou}}(S,G) = 1 - \frac{|S\cap G|}{|S|+|G|-|S\cap G|} \quad (15)$$

where S and G represent the saliency region in the prediction image and the GT, respectively. $\cap$ represents the intersection operation.

COS is originally used to evaluate the similarity between two vectors [60]. Drawing on the similarity evaluation, we introduce the COS loss for our method. The COS loss is

$$L_{\cos}(S,G) = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}(G_{ij}S_{ij})}{\sqrt{\sum_{i=1}^{H}\sum_{j=1}^{W}G_{ij}^2}\sqrt{\sum_{i=1}^{H}\sum_{j=1}^{W}S_{ij}^2}}. \quad (16)$$

It can be directly calculated with the prediction map and ground truth (GT). We can see that when the two images are more similar, the COS value is higher, and the COS loss is lower. When two images are identical, the COS value is 1 and the cosine loss is 0. Therefore, it can supervise the similarity of the overall pixels and maintain the accuracy of the global pixels.

The proposed novel loss is made up of the BCE loss, IoU loss, and COS loss, which is formulated as

$$L(S,G) = L_{\text{bce}}(S,G) + L_{\text{iou}}(S,G) + L_{\cos}(S,G). \quad (17)$$

IV. Experiments

We show and discuss some experimental results of the ACNet. First, the datasets and evaluation indicators are presented. Next, the implementation details are clearly introduced. Then, we show the visual and quantitative results of our ACNet and other methods. The comparison results demonstrate that the ACNet outperforms the 15 SOTA approaches. Finally, the ablation experiments and feature visualization further demonstrate that the designed modules and our hybrid loss in ACNet are effective.

A. Datasets

The experiments are conducted on two public optical RSI-SOD datasets including ORSSD [43] and EORSSD [44]. ORSSD [43] consists of 800 optical RSIs and corresponding GT, including diverse scenes, multiple scales, and various types of objects. The official division of the dataset is 600 training images and 200 testing images. EORSSD [44] is the extended dataset of ORSSD, which contains 2000 optical RSIs and corresponding GT. The EORSSD dataset contains more complex and difficult scenarios than the ORSSD dataset. The official division of the dataset is 1400 training images and 600 testing images.

To augment our training dataset, we conduct rotation (90°, 180°, and 270°) and execute mirrors of the images on the training sets of ORSSD and EORSSD. In this way, our training set contains 16 000 images, including 4800 images from ORSSD and 11 200 images from EORSSD. For fair comparisons, in the training phase, we train our ACNet and the compared methods on the same augmented training images. And the results of the comparison methods are obtained according to the recommended network and hyperparameters. In the test phase, we evaluate and compare the performance of all methods on the testing sets of ORSSD and EORSSD.

B. Evaluation Metrics

To objectively represent the performance of all methods, we adopt eight quantitative indicators including mean absolute error (MAE), (max, and mean) F-measure (F_β), (max, and mean) E-measure (E_ξ) [61], S-measure (S_α) [62], precision–recall (PR) curve, and F-measure curve.

MAE evaluates the average pixel difference between the prediction and the GT. The formula of MAE is

$$\text{MAE} = \frac{1}{W\times H}\sum_{x=1}^{W}\sum_{y=1}^{H}|S(x,y)-G(x,y)| \quad (18)$$

where W and H represent the width and height of the images.

F-measure is a comprehensive evaluation that combines precision and recall. Its formula is

$$F_\beta = \frac{(1+\beta^2)\times \text{Precision}\times \text{Recall}}{\beta^2\times \text{Precision} + \text{Recall}} \quad (19)$$

where $\beta^2 = 3$ is to emphasize the precision [63].

The formulas of precision and recall are

$$\text{Precision} = \frac{|S\cap G|}{|S|}, \quad \text{Recall} = \frac{|S\cap G|}{|G|} \quad (20)$$

where $\cap$ represents the operation on obtaining the intersection region between S and G.

We binarize the saliency prediction images through different thresholds. Then, the precision–recall pairs are obtained through the binarized result and GT. After the above operations, PR and F-measure curves can be plotted.

E-measure evaluates the local matching information on pixels and global statistics on images. The formula of E-measure is

$$E_\xi = \frac{1}{W\times H}\sum_{x=1}^{W}\sum_{y=1}^{H} f\left(\frac{2\varphi_{\text{GT}}\circ\varphi_{\text{FM}}}{\varphi_{\text{GT}}\circ\varphi_{\text{GT}}+\varphi_{\text{FM}}\circ\varphi_{\text{FM}}}\right) \quad (21)$$

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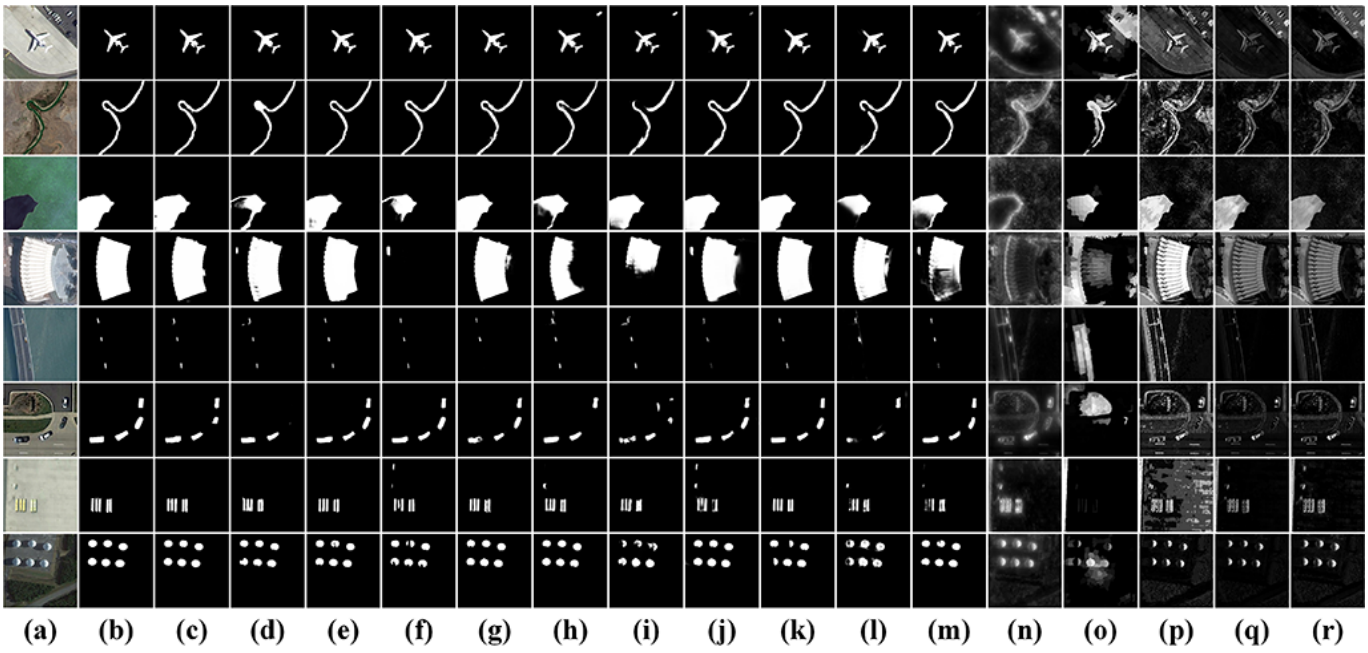


Fig. 3. Visual comparison between our ACNet and 15 SOTA methods. (a) Optical RSIs. (b) GT. (c) Ours. (d) FSMINet [53]. (e) ERPNet [51]. (f) CorrNet [52]. (g) RRNet [27]. (h) DAFNet [44]. (i) LVNet [43]. (j) GCPANet [34]. (k) BASNet [24]. (l) PoolNet [33]. (m) CPD [23]. (n) SMFF [42]. (o) CMC [41]. (p) VOS [40]. (q) FT [31]. (r) LC [30].

where $f()$ is a quadratic function and $\circ$ denotes the Hadamard product. φ denotes the bias matrix, which measures the distance between each pixel-wise value of GT and the image-level average. φ_{GT} and φ_{FM} are for GT and foreground map.

S-measure can estimate the structural similarity in SOD, which mainly focuses on the structural similarities of object-aware (S_o) and region-aware (S_r). Its formula is

$$S_\alpha = \alpha \times S_o + (1-\alpha) \times S_r \tag{22}$$

where the trade-off parameter α is 0.5 as suggested.

C. Implementation Details

We complete the experiments with PyTorch 1.9.0 on NVIDIA GeForce RTX 3090 GPU (24 GB memory). We train our ACNet and the compared methods on the augmented training images. The input and output are resized to 256×256. The ACNet is trained with the Adam algorithm [64]. The batch size is 8 and the initial learning rate is 3×10^{-5}. The ACNet trains 150 epochs with the hybrid supervision. Finally, we save the optimal model and generate saliency maps.

D. Comparative Experiments

We present some visual and quantitative comparison results, and the experimental results show the performances. Specifically, our ACNet is compared with 15 SOTA methods, including six CNN-based methods for optical RSIs (FSMINet [53], ERPNet [51], CorrNet [52], RRNet [27], DAFNet [44], and LVNet [43]), three traditional methods for optical RSIs (VOS [40], CMC [41], and SMFF [42]), four CNN-based methods for NSIs (GCPANet [34], BASNet [24], PoolNet [33], and CPD [23]), and two traditional methods for NSIs (FT [31], LC [30]).

1) Visual Comparison: We compare our ACNet with 15 SOTA methods under different scenes. Fig. 3 displays visual comparisons. We can see that the scenes mainly contain the single small object (first row), single irregular object (second row), single large object (third row), single object with interference (fourth row), multiple tiny objects (fifth row), multiple objects with interference (sixth row), multiple objects with low contrast (seventh row), and multiple objects with shadows (the last row).

Specifically, our ACNet can detect the single small object and completely retain the structure and details, while the SOTA methods detect the incomplete and fuzzy structures of the object. For the single irregular object, our ACNet can accurately peel off the correct irregular object and retain its sharp details, while some of the SOTA methods cannot locate the correct region, and the details are discontinuous. For the single large object, our ACNet can maintain the region and structure complete. For the single object with interference, the SOTA methods omit some important regions, and their contour and details are inaccurate. On the contrary, the region of our prediction is coherent and the details are sharp. For the multiple tiny objects, our ACNet can distinguish multiple objects and preserve the clear edges. For the multiple objects with interference, our ACNet does not miss any salient objects. For the multiple objects with low contrast, our method can detect multiple objects and maintain clear contours. For the multiple objects with shadows, the SOTA methods often ignore the shapes of the objects. On the contrary, our method can distinguish multiple objects and maintain their shapes. Accordingly, our ACNet can locate the accurate saliency region and retain abundant details under complex scenes and various scales. The visual comparisons present that our ACNet outperforms other SOTA methods under different scenes.

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TABLE I

QUANTITATIVE COMPARISON RESULTS BETWEEN OUR ACNET AND 15 SOTA METHODS ON TWO PUBLIC OPTICAL RSI-SOD DATASETS. "↑" INDICATES THAT THE HIGHER VALUES ARE BETTER, AND "↓" INDICATES THAT THE LOWER VALUES ARE BETTER. THE BEST THREE RESULTS ARE LABELED IN RED, GREEN, AND BLUE

Methods	FLOPs (G)↓	Params (M)↓	ORSSD [43]						EORSSD [44]					
			MAE↓	maxF_β↑	meanF_β↑	maxE_ξ↑	meanE_ξ↑	S_α↑	MAE↓	maxF_β↑	meanF_β↑	maxE_ξ↑	meanE_ξ↑	S_α↑
LC [30]	-	-	0.1228	0.4245	0.3111	0.7257	0.5490	0.5911	0.0855	0.4585	0.3330	0.7104	0.5591	0.5947
FT [31]	-	-	0.1123	0.4600	0.3094	0.7343	0.5303	0.5900	0.0710	0.4807	0.3378	0.7096	0.5462	0.6077
VOS [40]	-	-	0.2151	0.3541	0.2780	0.6514	0.5369	0.5366	0.2096	0.2862	0.2187	0.5999	0.4910	0.5083
CMC [41]	-	-	0.1267	0.3993	0.3592	0.7064	0.6424	0.6036	0.1057	0.3373	0.2784	0.6818	0.5908	0.5798
SMFF [42]	-	-	0.1854	0.4507	0.2747	0.7402	0.4933	0.5313	0.1434	0.5368	0.3116	0.7768	0.5225	0.5407
CPD [23]	9.38	47.85	0.0131	0.9047	0.8816	0.9691	0.9444	0.9232	0.0101	0.8565	0.8169	0.9655	0.9178	0.9022
PoolNet [33]	53.63	123.40	0.0237	0.8261	0.7996	0.9218	0.8955	0.8681	0.0147	0.7895	0.7506	0.9094	0.8681	0.8538
BASNet [24]	127.40	87.06	0.0130	0.9166	0.8934	0.9679	0.9509	0.9278	0.0079	0.8986	0.8558	0.9626	0.9375	0.9231
GCPANet [34]	67.06	54.30	0.0131	0.9080	0.8722	0.9675	0.9406	0.9173	0.0093	0.8540	0.8008	0.9460	0.9026	0.8932
LVNet [43]	-	-	0.0207	0.8432	0.8162	0.9456	0.9259	0.8813	0.0145	0.8065	0.7591	0.9279	0.8828	0.8650
DAFNet [44]	68.51	29.35	0.0144	0.9025	0.8557	0.9672	0.9316	0.9036	0.0101	0.8700	0.7903	0.9576	0.8946	0.8834
RRNet [27]	172.33	68.98	0.0129	0.9088	0.8827	0.9657	0.9474	0.9264	0.0085	0.8870	0.8431	0.9608	0.9309	0.9181
CorrNet [52]	21.09	4.09	0.0110	0.9213	0.9126	0.9711	0.9666	0.9352	0.0082	0.8996	0.8844	0.9654	0.9605	0.9271
ERPNet [51]	171.76	77.19	0.0120	0.9250	0.9040	0.9717	0.9566	0.9340	0.0082	0.9000	0.8635	0.9646	0.9407	0.9258
FSMINet [53]	5.24	3.56	0.0113	0.9135	0.9005	0.9727	0.9637	0.9301	0.0079	0.8943	0.8750	0.9681	0.9578	0.9276
Ours	122.26	57.98	0.0100	0.9238	0.9067	0.9764	0.9673	0.9382	0.0068	0.9060	0.8771	0.9719	0.9596	0.9330

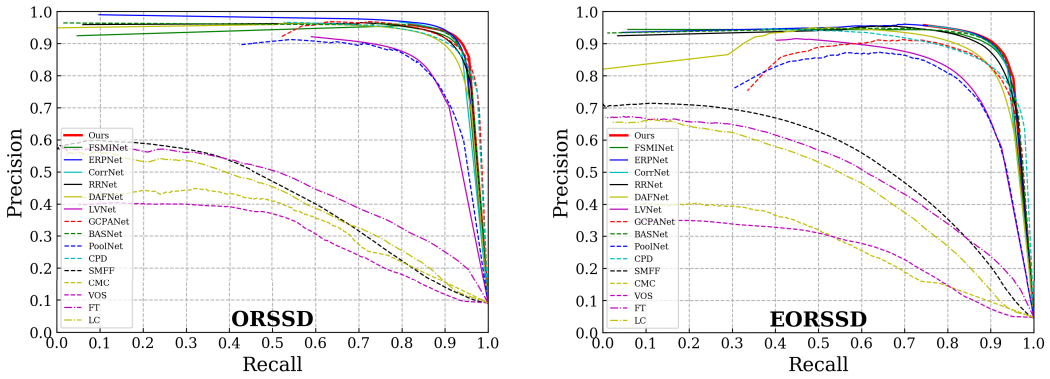
2) Quantitative Comparison: Table I shows the quantitative comparison results between our ACNet and 15 SOTA methods on two public optical RSI-SOD datasets in MAE, F_β, E_ξ, and S_α. Specifically, on the ORSSD dataset, our ACNet gets the highest scores in MAE, max E_ξ, mean E_ξ, and S_α. And our ACNet gets the second-best scores in max F_β and mean F_β. On the EORSSD dataset, our ACNet gets the highest scores in MAE, max F_β, max E_ξ, and S_α. And our ACNet gets the second-best scores in mean F_β and mean E_ξ.

From the quantitative results, we can conclude that our ACNet performs great and has strong competitiveness with SOTA methods. Table I also presents the efficiency of the different methods in terms of the floating point operations (FLOPs) and the number of parameters (Params). Compared with the other methods, the FLOPs and Params are medium. Therefore, the FLOPs and Params of our ACNet are reasonable, and the performance is great. In addition, Fig. 4 shows the PR curves of all methods on two public datasets. We can observe that the PR curve of our ACNet is closer to the upper right corner than the other SOTA methods. Fig. 5 shows the F-measure curves of different methods on two public datasets. The prediction images are grayscale images, and their value is within the range of [0, 255]. So, the threshold used to plot the F-measure curves is within the range of [0, 255]. We can observe that the F-measure curve of our ACNet is higher than the other SOTA methods. The results of the F-measure and PR curve present our ACNet performs better. In conclusion, these quantitative results intuitively illustrate that our ACNet is more robust and competitive.

E. Ablation Studies

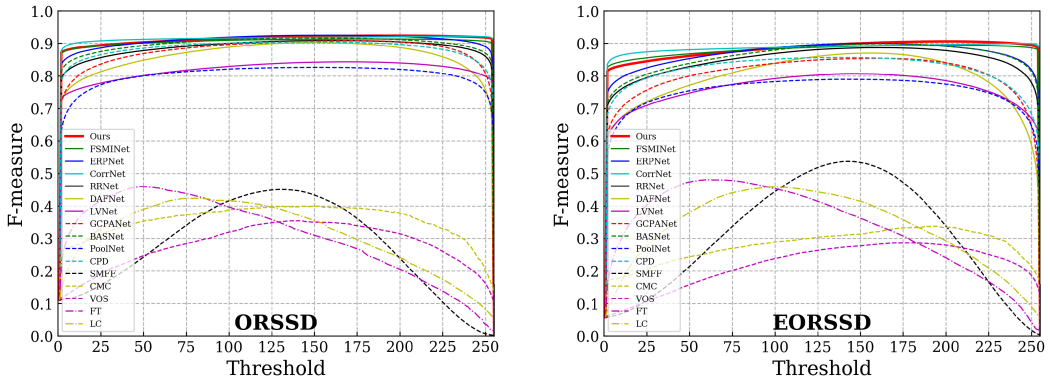
1) Effectiveness of Each Module in ACNet: To intuitively present the effectiveness of each designed modules in ACNet, we carry out quantitative ablation studies. The quantitative experiment results on two datasets are shown in Table II. The "Baseline" is an end-to-end network, and its encoder backbone is the modified VGG. And we use the BCE loss to train the "Baseline." "+ ACM" means the addition of the ACM to "Baseline," which illustrates the importance of the ACM. "+ GAP" means the addition of the GAP to "+ ACM," which illustrates the importance of the GAP. "+ FEM" means the addition of the FEM to "+ GAP," which illustrates the importance of the FEM. "+ Hybrid Loss" represents our final ACNet and means the addition of our hybrid loss to "+ FEM," which illustrates the importance of our hybrid loss. On the ORSSD dataset, the "Baseline" only achieves 0.0171 in MAE, 0.8954 in max F_β, and 0.9506 in max E_ξ. We can observe that the "+ ACM" improves the performance most and can improve "Baseline" by 0.38%, 1.09%, and 0.98% in MAE, max F_β, and max E_ξ. "+ GAP" and "+ FEM" slightly improve the performance. "+ Hybrid Loss" greatly improves the performance, and the ACNet can improve "Baseline" by 0.71%, 2.84%, and 2.58% in MAE, max F_β, and max E_ξ. On the EORSSD dataset, the "Baseline" only achieves 0.0099 in MAE, 0.8821 in max F_β, and 0.9499 in max E_ξ. We can also observe that the "+ ACM" improves the performance most and can improve "Baseline" by 0.12%, 1.37%, and 1.36% on MAE, max F_β, and max E_ξ. "+ GAP" and "+ FEM" slightly improve the performance. "+ Hybrid Loss" greatly improves the performance, and the ACNet can

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(a) (b)

Fig. 4. PR curves of different methods on two public datasets. (a) ORSSD [43]. (b) EORSSD [44].



(a) (b)

Fig. 5. F-measure curves of different methods on two public datasets. (a) ORSSD [43]. (b) EORSSD [44].

TABLE II

QUANTITATIVE ABLATION STUDIES ON EVALUATING EACH MODULE IN ACNET. THE BEST RESULT IS IN BOLD

Methods	ORSSD [43]			EORSSD [44]		
	MAE↓	maxF_β↑	maxE_ξ↑	MAE↓	maxF_β↑	maxE_ξ↑
Baseline	0.0171	0.8954	0.9506	0.0099	0.8821	0.9499
+ ACM	0.0133	0.9063	0.9604	0.0087	0.8958	0.9635
+ GAP	0.0132	0.9131	0.9657	0.0083	0.9001	0.9656
+ FEM	0.0120	0.9174	0.9707	0.0082	0.8997	0.9674
+ Hybrid Loss	**0.0100**	**0.9238**	**0.9764**	**0.0068**	**0.9060**	**0.9719**

TABLE III

QUANTITATIVE ABLATION STUDIES ON DESIGNED ACM. THE BEST RESULT IS IN BOLD

Methods	ORSSD [43]			EORSSD [44]		
	MAE↓	maxF_β↑	maxE_ξ↑	MAE↓	maxF_β↑	maxE_ξ↑
w/o ACM	0.0149	0.9040	0.9565	0.0089	0.8901	0.9568
Ours	**0.0100**	**0.9238**	**0.9764**	**0.0068**	**0.9060**	**0.9719**

improve "Baseline" by 0.31%, 2.39%, and 2.20% in MAE, max F_β, and max E_ξ. The quantitative ablation results reveal the performance of each module in ACNet. We can conclude that each module in ACNet is effective.

To individually demonstrate the effectiveness of the designed ACM, we carry out quantitative ablation studies. The results on two datasets are shown in Table III. w/o ACM means ACNet without ACM. We can see that the performance of ACNet without ACM has decreased a lot, so the ACM module is very useful to aggregate adjacent features.

2) Effectiveness of the Hybrid Loss: To demonstrate the effectiveness of the novel hybrid loss, we carry out quantitative ablation studies, which are shown in Table IV. First, we respectively use BCE loss, IoU loss, and COS loss to train the proposed ACNet, but their performances are fairly poor. Then, we respectively use the BCE loss with IoU loss, the BCE loss with COS loss, and the IoU loss with COS loss to train the proposed ACNet. Although their performances have been greatly improved, the results are still not satisfactory. Finally, our novel loss combined with BCE loss, IoU loss, and COS loss achieves the best performance. Compared with the single BCE loss, our hybrid loss improves the performance by 0.20%, 1.35%, and 1.08% in MAE, mean F_β, and mean E_ξ on the ORSSD dataset and improves the performance by 0.14%, 2.16%, and 1.75% in MAE, mean F_β, and mean E_ξ on the EORSSD dataset.

The visual results of our hybrid loss are shown in Fig. 6. We can see that the result in Fig. 6(c) is complete and continuous, the results in Fig. 6(d) and (g) omit some saliency regions, the result in Fig. 6(e) contains discontinuous

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TABLE IV
QUANTITATIVE ABLATION STUDIES ON THE HYBRID LOSS, THE BEST RESULT IS IN BOLD

BCE	IoU	COS	ORSSD [43]			EORSSD [44]		
			MAE↓	meanF_β↑	meanE_ξ↑	MAE↓	meanF_β↑	meanE_ξ↑
✓			0.0120	0.8932	0.9565	0.0082	0.8555	0.9421
	✓		0.0127	0.8966	0.9572	0.0085	0.8713	0.9540
		✓	0.0134	0.8924	0.9544	0.0090	0.8562	0.9402
✓	✓		0.0119	0.9035	0.9639	0.0077	0.8757	0.9593
✓		✓	0.0115	0.8981	0.9619	0.0078	0.8624	0.9466
	✓	✓	0.0116	0.8989	0.9629	0.0077	0.8747	0.9587
✓	✓	✓	**0.0100**	**0.9067**	**0.9673**	**0.0068**	**0.8771**	**0.9596**

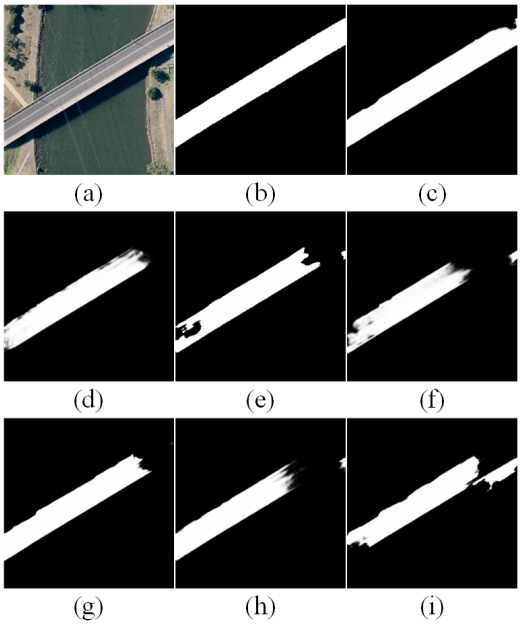


Fig. 6. Visual results on the proposed loss. (a) Optical RSIs. (b) GT. The result of (c) our novel loss, (d) BCE loss, (e) IoU loss, (f) COS loss, (g) BCE loss with IoU loss, (h) BCE loss with COS loss, and (i) IoU loss with COS loss.

regions, the results in Fig. 6(f) and (h) contain blurred boundaries, the result in Fig. 6(i) contains inaccurate regions. Therefore, it can be seen from these results that the proposed loss can more accurately preserve the structure and details.

According to the above results and analysis, the novel loss can supervise the similarity of overall pixels and maintain the complete objects.

3) Statistical Significance Analysis: We have shown the results of the statistical significance analysis and repeated our method five times to demonstrate its effectiveness and robustness. The results of our ACNet are shown in Table V. From the results, we can see that the standard deviation value is small, which means the performance is stable. Therefore, our ACNet performs effectively and robustly under repeated implementations.

TABLE V
RESULTS OF STATISTICAL SIGNIFICANCE ANALYSIS, INCLUDING MEAN (MEAN), AND STANDARD DEVIATION (STD)

Repeat	ORSSD [43]			EORSSD [44]		
Times	MAE↓	meanF_β↑	meanE_ξ↑	MAE↓	meanF_β↑	meanE_ξ↑
1	0.0100	0.9067	0.9673	0.0068	0.8771	0.9596
2	0.0106	0.9078	0.9670	0.0073	0.8790	0.9609
3	0.0106	0.9123	0.9682	0.0071	0.8778	0.9593
4	0.0101	0.9059	0.9648	0.0073	0.8733	0.9582
5	0.0101	0.9112	0.9706	0.0073	0.8763	0.9600
Mean	0.0103	0.9088	0.9676	0.0072	0.8767	0.9596
Std	0.0003	0.0028	0.0021	0.0002	0.0021	0.0010

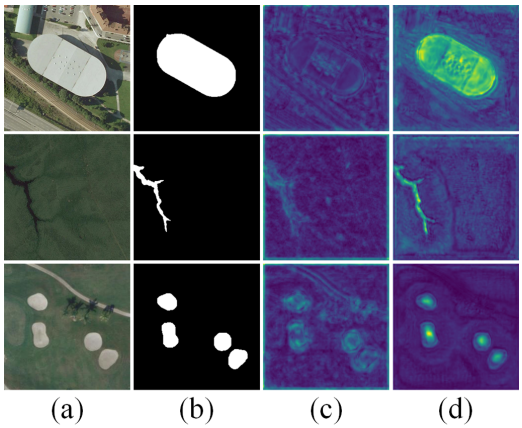


Fig. 7. Feature visualization of ACM and FEM. (a) Optical RSIs. (b) GT. Feature visualization from (c) $\boldsymbol{ACM_2}$ (f_{ACM_2}) and (d) $\boldsymbol{FEM_2}$ (f_{D_2}).

F. Feature Visualization of Proposed ACNet

The ACM and FEM are the two main modules in ACNet. To make our ACNet more convincing, we display the feature visualizations from the proposed ACM and FEM, which are shown in Fig. 7. We can see that the ACM can effectively capture saliency features, and the FEM can highlight saliency regions. Therefore, our ACNet can accurately detect salient objects in various scenes.

V. CONCLUSION

In this article, we propose the ACNet with global constraints for optical RSI-SOD. The ACNet is designed to alleviate the issue of insufficient adjacent feature representation and supervision constraints. To improve the detection performance of objects with complex backgrounds and various scales, our ACNet is specially equipped with the ACM, GAP, FEM, and novel loss. In the encoder, the ACM is used to fuse and complement the features among the current layer and adjacent layers, which can effectively aggregate saliency features. Then, the GAP is equipped in the last layer of the encoder, which can enhance global information. In the decoder, the FEM can enhance the feature conversion between the coding layer and decoding layer. Finally, the novel loss introduces the global constraints, which can globally supervise objects with various scales. Adequate experiments have presented the superiority and effectiveness of our ACNet.

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